Quantification: 682_z08
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	10
interior lacuna count	8
frame-edge lacuna count	2

Interior lacunae only: mean, median and sample SD (n = 8)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2599.88	2671.5	1127.64	px^2
major_axis_length_px	lacuna shape	103.523	92.8874	29.9285	px
minor_axis_length_px	lacuna shape	33.6052	36.6597	6.7292	px
aspect_ratio	lacuna shape	3.087	2.9214	0.6438	unitless
eccentricity	lacuna shape	0.9401	0.9395	0.0221	unitless
solidity	lacuna shape	0.8296	0.873	0.1073	unitless
roots_count	headline	8.875	8	3.5632	count
ring_length_r30_px	headline	460	400.5	143.786	px
ring_length_r60_px	reported	1083.38	939.5	361.322	px
owned_length_px	ownership-dependent	2933.14	1646.03	2286.3	px
edge_count	ownership-dependent	138.125	82	105.084	count
mean_edge_length_px	ownership-dependent	21.3005	21.6347	1.7632	px
perimeter_px	normalised	275.426	240.43	89.7432	px
ring_area_r30_px2	normalised	10379.6	9533.5	2334.04	px^2
ring_area_r60_px2	normalised	24427.5	24429.5	5136.87	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9949	1	0.0144	unitless
ring_density_r30	normalised	0.0440793	0.0419509	0.0069352	px^-1
ring_density_r60	normalised	0.0440589	0.043665	0.00814855	px^-1
roots_per_100px_perimeter	normalised	3.2531	3.1586	0.9129	per 100 px
ring_attached_length_r30_px	network v2	397.75	336	138.692	px
ring_attached_length_r60_px	network v2	902.875	769	392.262	px
ring_length_w_r30_px	network v2	520.857	452.812	155.353	px
ring_length_w_r60_px	network v2	1226.08	1049.9	394.154	px
sholl_crossings_r10	network v2	10.875	9.5	3.9799	count
sholl_crossings_r20	network v2	14.5	13.5	3.7033	count
sholl_crossings_r30	network v2	16.25	13.5	5.3117	count
ring_width_mean_r30_px	width	7.1884	7.3449	0.4664	px

Per field
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measure	value	units
analysed_area_px2	1.01791e+06	px^2
total_skeleton_length_px	35082	px
canalicular_length_density_per_px	0.0344648	px^-1
junction_count	791	count
junction_density_per_px2	0.00077708	px^-2
lacuna_count	10	count
lacunae_per_px2	9.82e-06	px^-2
skeleton_component_count	279	count

mean_component_length_px	125.742	px
median_component_length_px	18	px
field_length_density_w_per_px	0.0388048	px^-1
field_density_without_flagged_per_px	0.0354417	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	1011	px
unattached_component_count	57	count
canalicular_length_density_attached_per_px	0.0344648	px^-1
field_length_density_with_unattached_per_px	0.035458	px^-1

Canalicular width, the new measure

measure	value	units
width_mean_px	7.5376	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10	px
width_pixels_used	33419	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	9321	210.577	69.7066	3.0209	0.9436	0.684	1.3112	31.8256	428.162	True
2	2563	90.1026	36.7546	2.4515	0.913	0.9144	1.5381	342.687	87.595	False
3	3496	113.895	40.712	2.7976	0.9339	0.8552	-1.2503	434.254	701.917	False
4	4123	159.751	36.5648	4.369	0.9735	0.7865	1.4743	433.966	210.1	False
5	1521	89.609	30.1271	2.9744	0.9418	0.641	-1.2077	520.501	556.951	False
6	862	61.0771	21.2942	2.8683	0.9373	0.7077	-1.4443	545.484	438.164	False
7	547	50.4784	17.0874	2.9541	0.941	0.8116	-0.4973	672.987	1013.45	True
8	3601	127.976	37.0473	3.4544	0.9572	0.8909	1.4857	713.926	152.006	False
9	1853	92.0913	27.1748	3.3889	0.9555	0.9155	-1.2655	723.731	828.852	False
10	2780	93.6835	39.1668	2.3919	0.9084	0.926	-1.1692	908.613	727.092	False

Network per lacuna

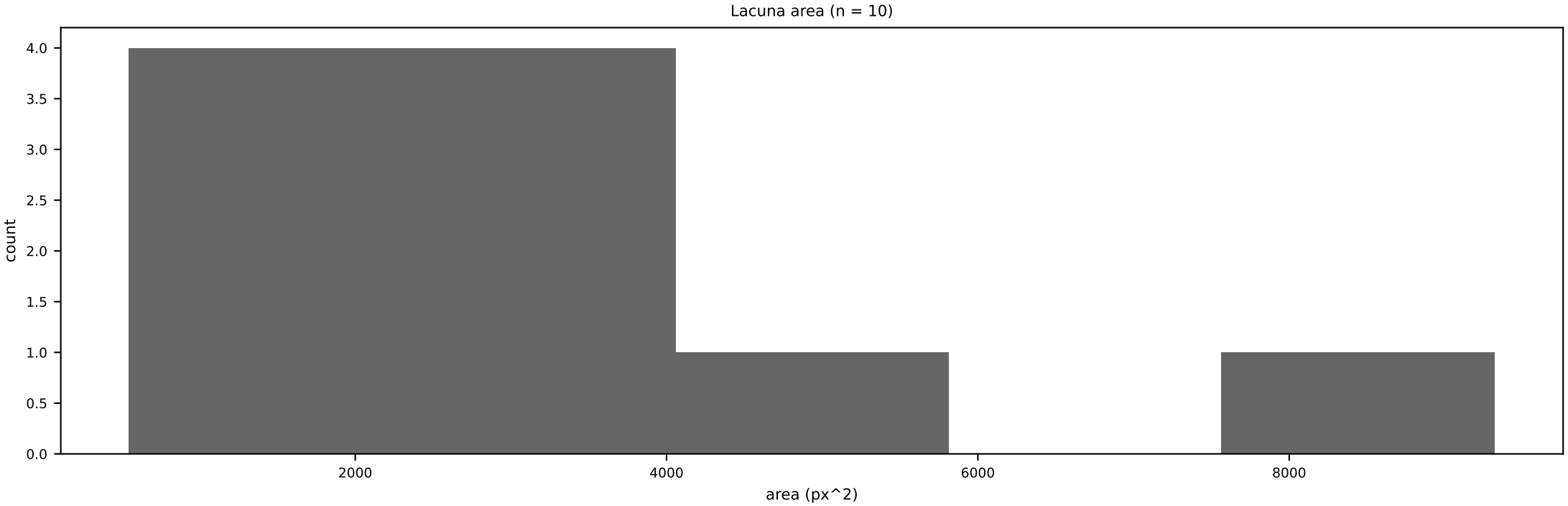
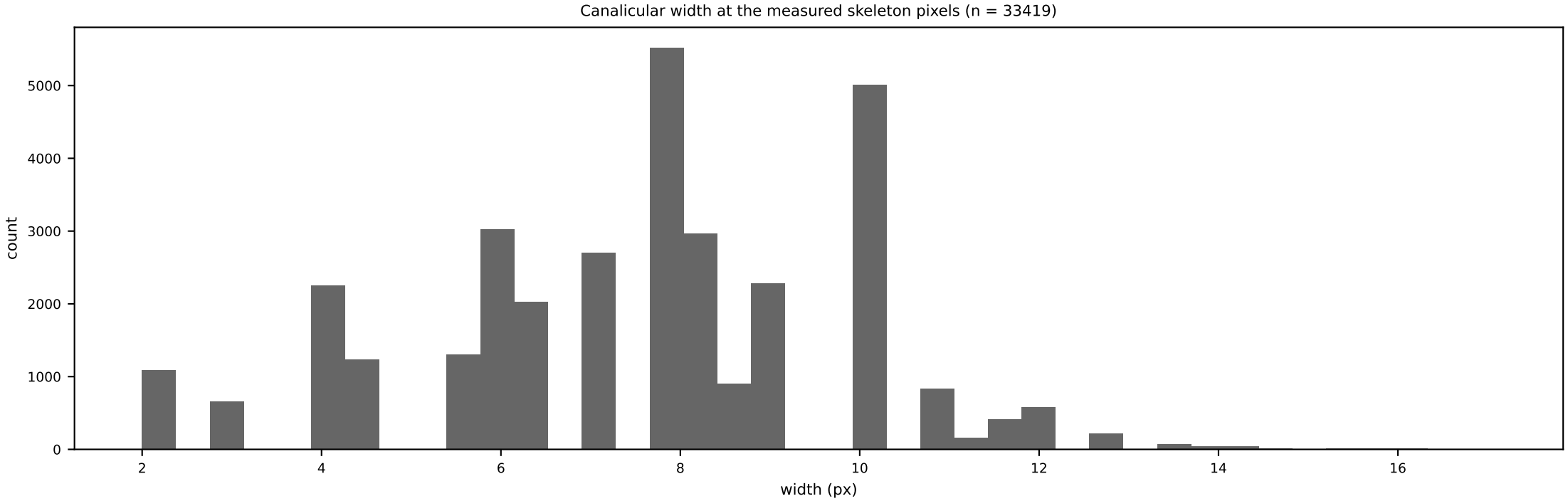
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	10	403	1050	4607.84	234	19.6916	5.308
2	6	378	961	1407.36	72	19.5467	7.5912
3	13	453	1087	3571.23	163	21.9094	7.4864
4	15	743	1814	5105.98	263	19.4144	7.6399
5	6	401	801	1580.64	74	21.36	6.8902
6	8	400	918	1711.43	90	19.0158	6.4357
7	3	144	402	405.044	12	33.7536	6.0372
8	10	614	1436	7460.37	330	22.6072	7.5804
9	8	366	833	1122.86	48	23.3929	7.2034
10	5	325	817	1505.24	65	23.1575	6.6802

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter

1	769.541	14367	29344	0.6579	0.6229	0.0280504	0.0357824	1.2995
2	234.894	9374	20769	1	0.9593	0.0403243	0.0462709	2.5543
3	313.948	11274	26474	1	1	0.0401809	0.0410592	4.1408
4	456.961	14983	32605	1	1	0.0495895	0.0556356	3.2826
5	233.238	9202	19621	1	1	0.0435775	0.0408236	2.5725
6	166.125	7387	17424	1	1	0.0541492	0.052686	4.8157
7	122.604	3917	10590	0.6212	0.5802	0.0367628	0.0379603	2.4469
8	329.522	12020	29668	1	1	0.0510815	0.0484023	3.0347
9	222.752	9104	23858	1	1	0.0402021	0.0349149	3.5914
10	245.965	9693	25001	1	1	0.0335293	0.0326787	2.0328

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	320	687	444.363	1138.68	14	13	12
2	322	853	425.877	1072.35	10	11	14
3	434	883	519.174	1247.71	15	13	13
4	671	1721	822.914	2034.83	17	22	27
5	350	685	465.375	930.921	6	14	13
6	319	667	440.25	1027.45	7	12	13
7	94	222	170.167	477.428	3	6	7
8	525	1225	691.701	1590.02	14	18	22
9	300	657	424.031	972.288	9	14	15
10	261	532	377.534	933.021	9	12	13

682_z08: distributions, pre-validation, px